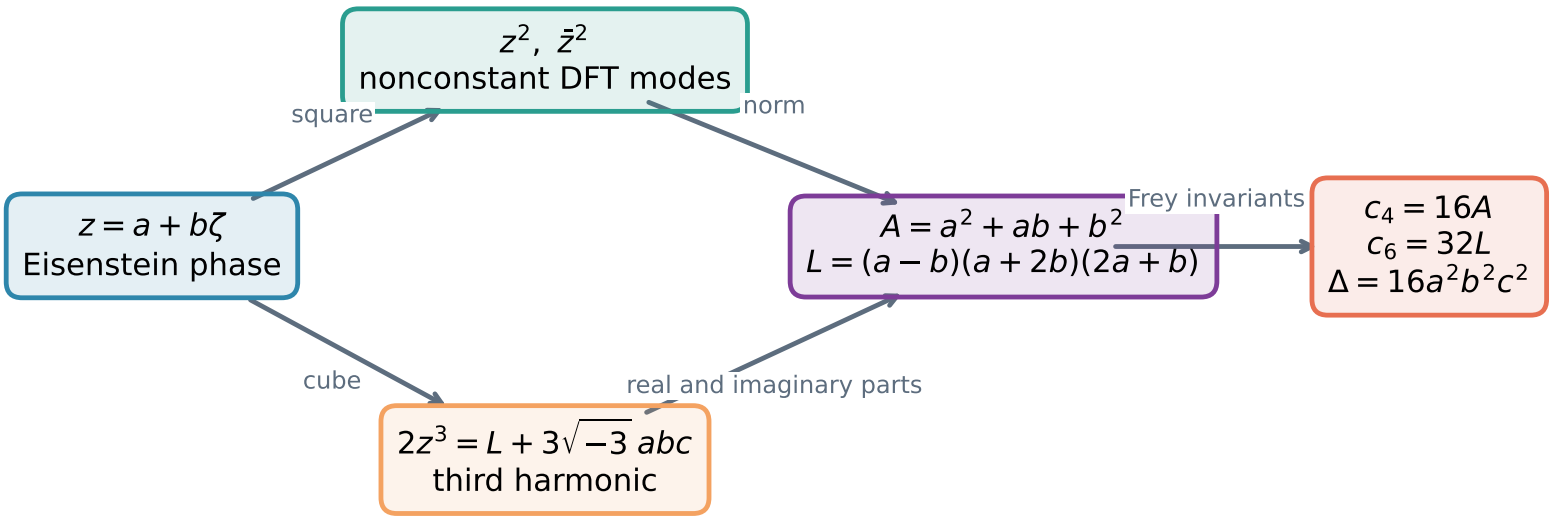The Frey invariants appear as harmonics of one phase packet